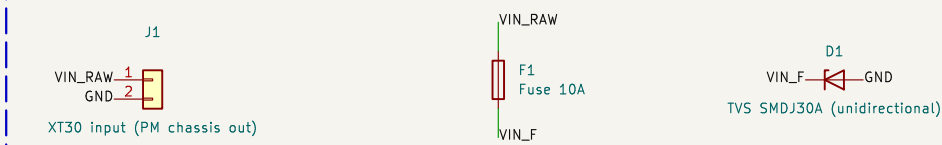


Rev B control board: block 1 / charger (input protection, series PMOS, LM5176 4-switch buck-boost) — plan Y, PROVISIONAL

Values: docs/revb/plan_y_charger_sizing.md. Connectivity by net labels. Verify against the original datasheet: ISNS+/- polarity, MODE (hiccup), BIAS=NODE, UVLO values, FET/inductor footprints.

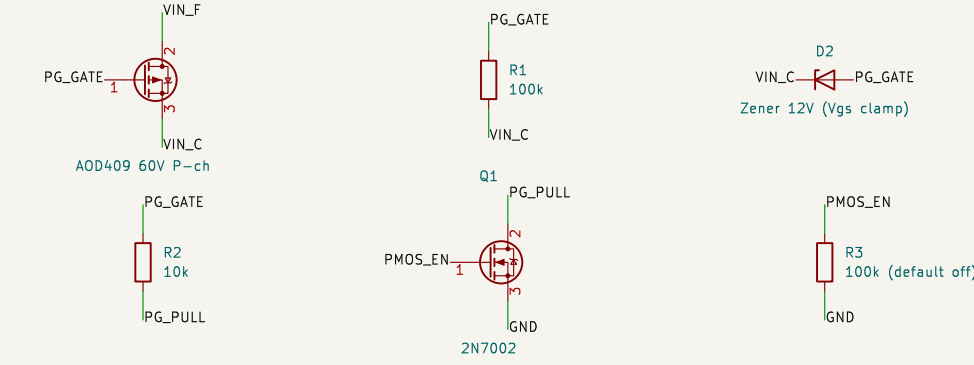
A. Input / protection

PM chassis output 22-26V. Fuse + TVS (SMDJ30A).



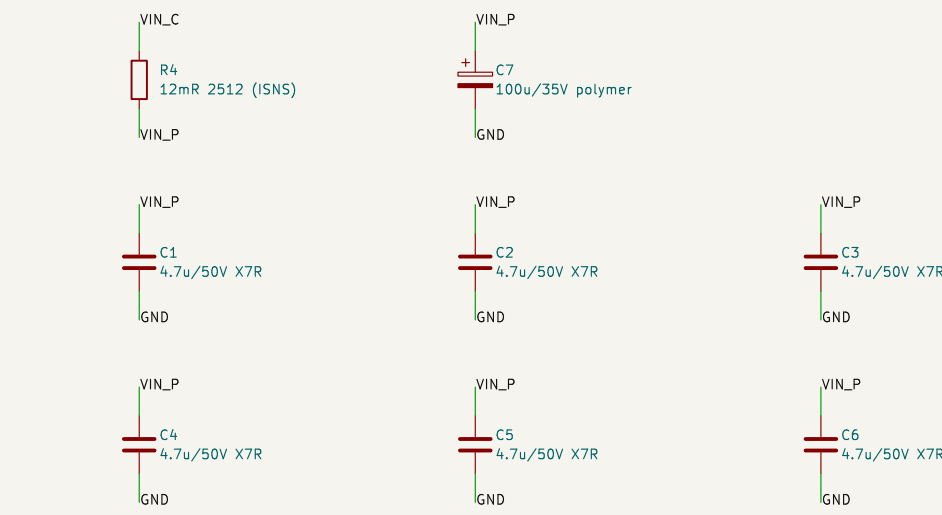
B. Series PMOS (R2/R3 back-feed block)

Default OFF, ON only when PMOS_EN=1 (Zener limits Vgs to 12V).



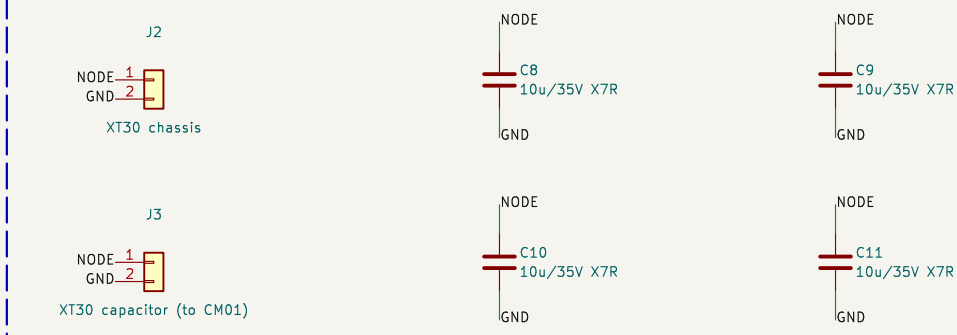
C. Input current sense / input capacitors

ISNS average limit 50mV/12mR = 4.2A (input power cap). Kelvin to ISNS+/-.



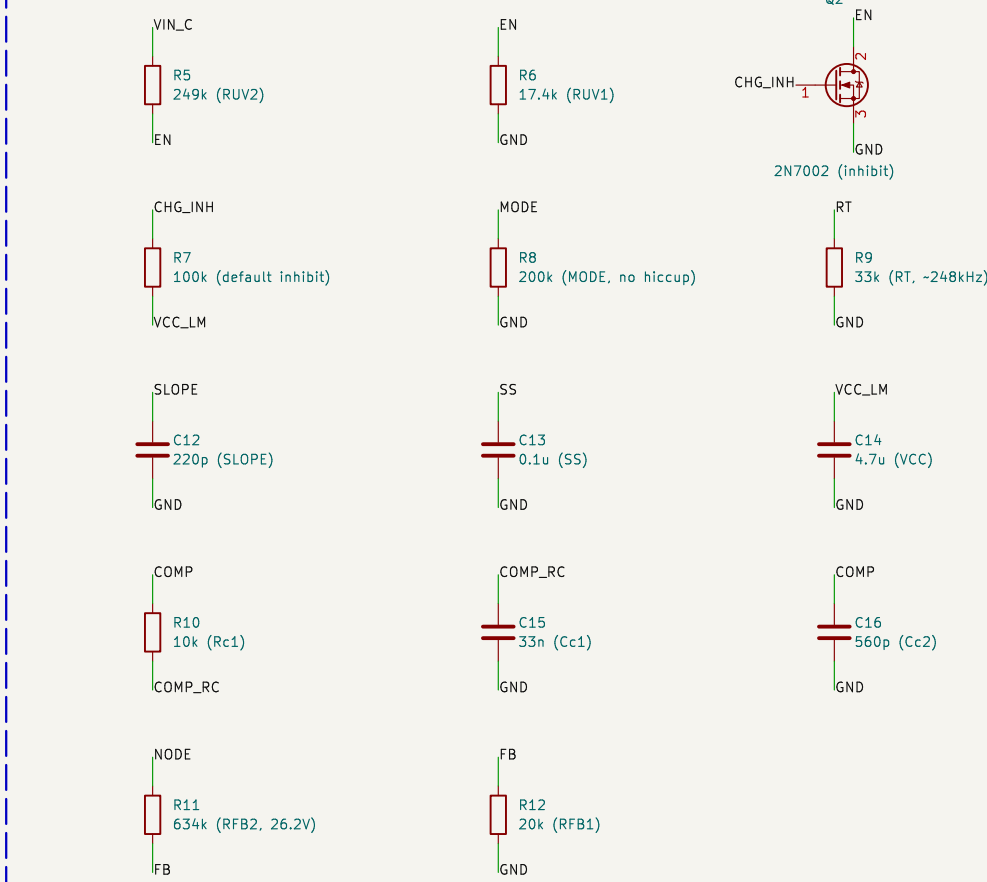
D. Output node (chassis + capacitor bank)

Chassis and bank share this node. Bank via CM01.



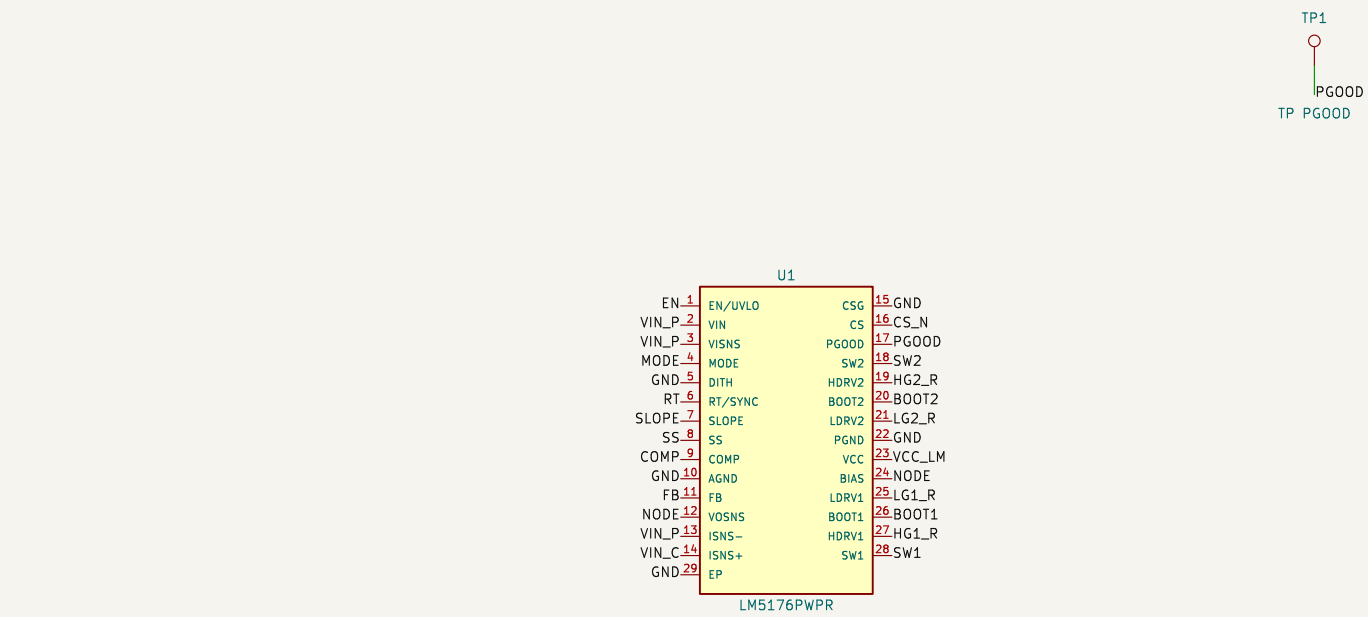
E. LM5176 settings

UVLO -19V, no hiccup, 248kHz, Vout 26.2V.



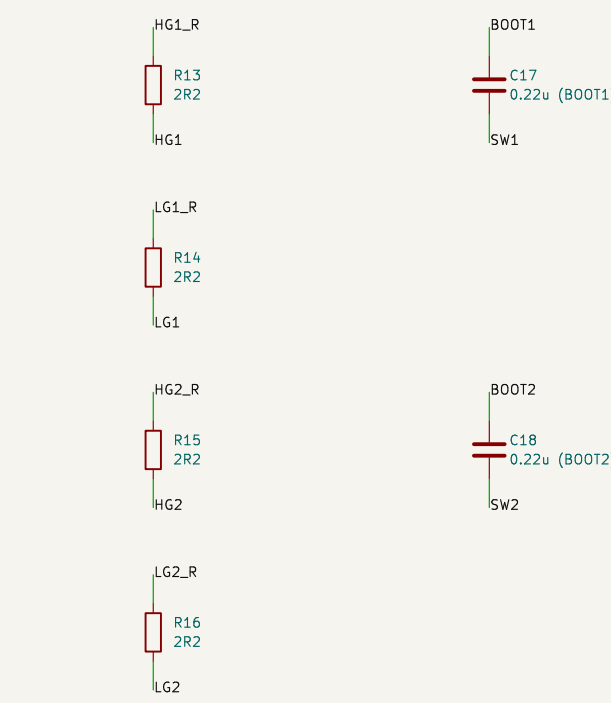
F. LM5176 controller

HTSSOP-28. Buck side = SW1/HDRV1/LDRV1, boost side = SW2/HDRV2/LDRV2.



G. Gate drive

Gate resistors and bootstrap capacitors.



H. Power stage

Q1,Q2 = buck leg; Q3,Q4 = boost leg; one Rcs for both low sides (CS/CSG Kelvin).

